

RESEARCH ARTICLE

Reduced-Complexity Belief Propagation Decoding for LDPC Codes

HEPING WAN¹, JOONYOUNG CHO¹, MIN JANG², (Member, IEEE),
AND CHARLIE JIANZHONG ZHANG¹, (Fellow, IEEE)

¹Samsung Research America, Plano, TX 75024, USA

²Samsung Research, Seoul 06765, South Korea

Corresponding author: Heping Wan (heping.wan@samsung.com)

ABSTRACT Two methods are proposed to reduce the complexity of the belief propagation (BP) decoding for low-density parity-check (LDPC) codes, with the focus on the layered decoding for quasi-cyclic LDPC (QC-LDPC) codes. The two methods can be separately incorporated into the layered decoding procedures and combined seamlessly to provide the two-fold complexity reduction. In the first method, the message updates regarding a subset of variable nodes are omitted. To construct the subset, the most reliable bit-level channels, derived from M -ary input channels, are identified. Then, we select the variable nodes with initial log-likelihood ratios (LLRs) exceeding a threshold among the ones assigned to the most reliable bit-level channels to form the subset. The threshold is designed and adjusted based on characteristics of the most reliable bit-level channels. Our numerical results demonstrate that the first method significantly reduces the complexity with negligibly impacting the error-correcting performance. In the second method, each layer of base graphs of QC-LDPC codes uses either sum-product algorithm (SPA) or min-sum algorithm (MSA) to update LLRs. A degree-based algorithm is presented to determine the layers employing either SPA or MSA. We compare with two intuitive methods: 1) applying SPA to randomly selected layers and MSA to the remaining layers, and 2) applying SPA to the core layers of 5G NR LDPC codes and MSA to the remaining layers. According to the numerical results, the degree-based algorithm maximally reduces the complexity while maintaining the same performance. To evaluate the joint method, the first method is applied on top of the second method. Further complexity reduction is achieved without degrading the performance of the second method.

INDEX TERMS LDPC codes, belief propagation decoding, reduced complexity.

I. INTRODUCTION

LDPC codes [1] are known for approaching channel capacity asymptotically, by using the BP decoding which allows for practical implementation and delivers high decoding throughput. As a result, they are adopted by industrial communication systems such as digital television broadcast via satellites (DVB-S2) and Wi-Fi, and are also employed for the data transmission in the 5th Generation (5G) New Radio (NR) standard [2], achieving a throughput of tens of gigabits per second.

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The BP decoding is essential for the successful implementation of LDPC codes. It is performed by iteratively passing messages within a Tanner graph representing LDPC codes. The graph consists of check nodes (CNs), variable nodes (VNs) and edges connecting them. The iterative message passing occurs between CNs and VNs which are neighbors, and continues until either the decoding converges or the maximum number of iterations is reached. As the convergence speed of the BP decoding increases, the number of iterations needed to obtain a valid codeword decreases, and vice versa.

Conventionally, the arrangement of updating messages depends on the flooding schedule, which is fulfilled by iterating the following three main steps: VNs report their

knowledge of bit variables to CNs *in parallel*; CNs combine the received information and pass back the extrinsic information *in parallel*; VNs update their knowledge *in parallel*, based on the extrinsic information from CNs. Although benefiting from full parallelism, the flooding schedule suffers from a slow convergence speed, because the updated knowledge are not utilized until next iteration. To speed up convergence, the BP decoding with the (row) layered schedule, commonly known as the layered decoding, was introduced in [3] and [4]. During each iteration, CNs are processed *sequentially*. When processing each CN, its neighboring VNs report their knowledge to the CN. Then, the CN returns new insights to the neighboring VNs for updating the knowledge. The sequential processing ensures the latest knowledge from VNs to be immediately exploited by the succeeding CNs. Sharon et al. [5] proved that the layered decoding converges twice as fast as the BP decoding with the flooding schedule, therefore reducing the necessary number of iterations by half [3].

In a Tanner graph, two CNs are considered disjoint if they do not share any common VNs. This property facilitates the partial parallelism in the layered decoding. Specifically, the CNs, which are mutually disjoint, can be processed in parallel without read-write conflicts. This partial parallelism can be efficiently realized for QC-LDPC codes [6]. CNs in a QC-LDPC Tanner graph can be divided into small number of groups and each group contains disjoint CNs. These groups are referred to as layers. In each layered decoding iteration, the layers are updated in order while the CNs of the same layer can be updated simultaneously.

The near-optimal performance of LDPC codes is guaranteed by SPA [1] which specifies the computations needed for updating messages. The accurate computations of the hyperbolic functions provided by SPA incur a significant increase in the computational complexity. By using its approximated versions, such as MSA [7], the normalized MSA [8] and the offset MSA [9], the computations are simplified at the cost of degradation in performance.

Other attempts to reduce the decoding complexity include skipping certain message calculations [10], [11], [12], [13], [14], [15]. Zimmermann et al. [10] proposed to detect the reliability of the VNs in each iteration and stop updating the reliable VNs in the subsequent iterations. The reliability depends on the comparison between the a posteriori LLRs (AP-LLRs) of the VNs and a threshold. This method is enhanced by employing an iteratively updated threshold [11], skipping LLR updates for the reliable CNs [12], [13], flipping wrong decisions [14] and deleting already converged VNs from the Tanner graph [15]. However, the above methods introduce extra computational complexity mainly due to the detection of the reliable VNs in each iteration.

This paper presents methods to reduce the complexity of the BP decoding. In the 5G NR and Wi-Fi standards, QC-LDPC codes are adopted to ensure the system reliability, with layered decoding commonly employed for implementation. Their superiority results from the aforementioned fast

convergence and (partially) parallel processing, alongside additional advantages like efficient encoding and flexible rate matching. This paper elaborates and validates the proposed methods based on the layered decoding for QC-LDPC codes.

The first method skips the LLR updates for a subset of VNs, referred to as the deactivated VNs (dVNs). The non-uniform reliability of bit-level channels in bit-interleaved coded modulation (BICM) systems is utilized to identify dVN. The identification depends on the characteristics of the initial LLRs. Therefore, the set of dVNs can be determined prior to performing the layered decoding and remain unchanged throughout the decoding iterations. An LLR threshold, obtained from the analysis of the probability density function (PDF) of the initial LLRs, is adjusted to manage the trade-off between the complexity and the error-correcting performance.

The second method benefits from both the precise LLR calculations and the low complexity, provided by SPA and MSA, respectively. Hybrid sum-product and min-sum (hSPMSA), where each layer employs either SPA or MSA, is proposed to achieve this goal. By analyzing the contributions of SPA and MSA to the complexity of hSPMSA, an algorithm for selecting the layers employing SPA and MSA is proposed to optimize the performance-complexity trade-off. Shortly, SPA is preferentially employed by the layers with less number of neighbors. Furthermore, the two methods can be integrated into the layered decoding together without any conflict, thereby obtaining the combined complexity reduction.

5G NR LDPC codes at a wide range of code rates and lengths are simulated to evaluate the proposed methods. The error-correcting performance and the decoding complexity are measured by the block error rate (BLER) and the number of float-point operations (FLOPs), respectively. By deactivating VNs, up to 33% of FLOPs can be saved with negligible impact on performance. In hSPMSA, the proposed algorithm for selecting the layers employing SPA and MSA achieves a 20% reduction in FLOPs compared to the random selection, while preserving the same level of performance.

II. PRELIMINARIES

A. NOTATION

$\mathbb{Z}_{i:j} \triangleq \{i, i+1, \dots, j\}$ represents the set of integers from i to j , with the abbreviation $\mathbb{Z}_j \triangleq \mathbb{Z}_{0:j}$ for convenience. $|\mathcal{A}|$ denotes the number of elements in a set $\mathcal{A}$. The inclusion and exclusion of an element a for the set $\mathcal{A}$ are represented as $\mathcal{A} + a$ and $\mathcal{A} - a$, respectively. $\mathcal{A} \setminus \mathcal{B}$ denotes the set of elements that are in the set $\mathcal{A}$ but not in the set $\mathcal{B}$. Given a matrix $\mathbf{M}$ of dimensions $m \times n$, we use $\mathbf{M}_{j,[0:n-1]}$, $\mathbf{M}_{[0:m-1],i}$ and $\mathbf{M}_{j,i}$ to represent the row j , the column i and the entries positioned at the row j and the column i , respectively.

B. LDPC CODES

An (N, K) LDPC code is defined by a (binary) parity check matrix $\mathbf{H}$ of dimensions $(N - K) \times N$, where N and K denote the code length and the information size, respectively. Each row in the parity check matrix represents a parity check

constraint. A valid codeword, denoted $\mathbf{v} \triangleq \{v_0, \dots, v_{N-1}\}$, must satisfy all parity check constraints, that says $\mathbf{H} \times \mathbf{v}^T = \mathbf{0}$.

The parity check matrix can be mapped to a bipartite graph, namely Tanner graph. It consists of N VNs and $(N - K)$ CNs. Additionally, there exists an edge connecting the CN j and the VN i if $H_{j,i}$ is non-zero. A CN and a VN are neighbors if they are connected. The degree of a CN is defined as the number of its neighboring VNs. The VN i corresponds to the codeword bit v_i , i.e., the column i of the parity check matrix. The CN j corresponds to the parity check constraint $\mathbf{H}_{j,[0:N-1]} \times \mathbf{v}^T = 0$, i.e., the row j of the parity check matrix. Messages (usually in the form of LLRs) are iteratively exchanged in a scheduled order between the neighboring nodes for the decoding purpose.

1) QC-LDPC CODES

The parity check matrix for $(N = n \times Z, K = k \times Z)$ QC-LDPC codes is defined by a $(n - k) \times n$ base graph:

$$\mathbf{B} = \begin{pmatrix} B_{0,0} & \cdots & B_{0,n-1} \\ \vdots & \ddots & \vdots \\ B_{n-k-1,0} & \cdots & B_{n-k-1,n-1} \end{pmatrix},$$

where Z is called the lifting size and the value of each entry belongs to $\mathbb{Z}_{-1:Z-1}$. By replacing each entry with a square matrix of size Z according to the specified lifting value, the base graph expands into the parity check matrix. More specifically, the entry is replaced by the all-zero matrix if its value is equal to -1 . Otherwise, it represents a permutation matrix obtained by circularly shifting the identity matrix to the right by the number of times equal to its value. By denoting the permutation matrix shifted $s \in \mathbb{Z}_{Z-1}$ times and the all-zero matrix as $\mathbf{I}^{(s)}$ and $\mathbf{I}^{(-1)}$, respectively, the parity check matrix of QC-LDPC codes is shown as:

$$\begin{pmatrix} \mathbf{I}^{(B_{0,0})} & \cdots & \mathbf{I}^{(B_{0,n-1})} \\ \vdots & \ddots & \vdots \\ \mathbf{I}^{(B_{n-k-1,0})} & \cdots & \mathbf{I}^{(B_{n-k-1,n-1})} \end{pmatrix} \begin{matrix} \leftarrow \text{Layer } 0 \\ \vdots \\ \leftarrow \text{Layer } n - k - 1 \end{matrix}.$$

The rows of the parity check matrix (equivalently the CNs in the Tanner graph) are naturally divided into $(n - k)$ groups referred to as *layers*. The circularly-shifting structure ensures the CNs belonging to the same layer are disjoint and have the same degrees.

2) 5G NR LDPC CODES

They belong to a class of QC-LDPC codes and underlie the evaluations of our study in Section V. Two base graphs, base graph 1 (BG1) and base graph 2 (BG2), are used to construct 5G NR LDPC codes [2]. Fig. 1a depicts BG2 which can be partitioned into sub-matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{O}$ and $\mathbf{I}$, where $\mathbf{B}$ and $\mathbf{I}$ are dual-diagonal and diagonal, respectively. Due to the structural similarities between BG1 and BG2, the depiction for BG1 is omitted. This structure facilitates the fast systematic encoding [16], [17]. The systematic bits,

comprising the sub-matrix $\mathbf{A}$, consist of information bits, CRC bits and shortening bits used for rate matching. Given the systematic bits $\mathbf{v}_1$, the precoded parity bits $\mathbf{v}_2$, comprising $\mathbf{B}$, are generated according to the constraint $\mathbf{A} \times \mathbf{v}_1^T = \mathbf{B} \times \mathbf{v}_2^T$. Then, the single parity check (SPC) bits, comprising $\mathbf{I}$, are generated according to the constraint $\mathbf{C} \times [\mathbf{v}_1, \mathbf{v}_2]^T = \mathbf{I} \times \mathbf{v}_3^T$. The resulting codeword is given by $[\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3]$.

BG2 consists of 42 rows and 52 columns. Given the systematic puncturing,¹ the minimum supported code rate is approximately $\frac{42-32}{52-2} \approx 0.2$. To support higher rates, we need to puncture the parity bits from the the end of the codewords. In the example shown in Fig. 1b, the shaded columns and rows due to the puncturing of 10 blocks of bits can be removed from BG2. Therefore, the resulting code rate is increased to $\frac{42-32}{42-2} \approx 0.25$, and the layered decoding can be performed over the sub-matrix which are not grayed out. We refer to the layers involved in decoding procedures as the *effective* layers. For example, the layers 0–31 are effective in Fig. 1b.

C. BITS TO SYMBOL MAPPING

The codeword bits of LDPC codes are partitioned into groups of $\log_2(M)$ bits when employing an M -ary modulation. Denote $v^{(i)}$ as the bit located at position $i \in \mathbb{Z}_{\log_2(M)-1}$ among $\log_2(M)$ bits. Then, each group $\{v^{(0)}, \dots, v^{(\log_2(M)-1)}\}$ is assigned to a transmitted symbol x belonging to a symbol set $\mathcal{X}$ of size M , via a bijective mapping.

D. LAYERED DECODING

Let $\mathcal{V}_j$ denote the set of indices of the VNs neighboring the CN j . For QC-LDPC codes, let ϕ_l denote the set of indices of the CNs belonging to the layer l . In the layered decoding, the following LLRs are updated iteratively:

- $\beta_{j \rightarrow i}$, the check-to-variable (C2V) LLR passed from the CN j to its neighboring VN i ,
- $\alpha_{i \rightarrow j}$, the variable-to-check (V2C) LLR passed from the VN i to its neighboring CN j ,
- γ_i , AP-LLR of the VN i .

At Step 1 of the Algorithm 1, AP-LLRs are initialized by initial LLRs $\{\lambda_i : i \in \mathbb{Z}_{N-1}\}$ obtained from channel observations. At Step 2, all C2V LLRs are initialized to zeros. $I_{\max}$ at Step 3 denotes the maximum number of iterations. During each iteration, the layers are processed in a predetermined order (see Step 4). The order is represented by $\pi(\{0, 1, \dots, n - k - 1\})$ which is a permutation of $\{0, 1, \dots, n - k - 1\}$. For each CN j belonging to the layer l , the LLRs associated with its neighboring VNs $\mathcal{V}_j$ are updated at Step 7 through Step 9, where the required computations depend on whether SPA or MSA is applied. The operation $\text{sgn}(\cdot)$ returns a value of minus one if the input argument is negative; otherwise, it returns a value of one.

¹The information bits corresponding to the first two columns of BG1 and BG2 are punctured [2].

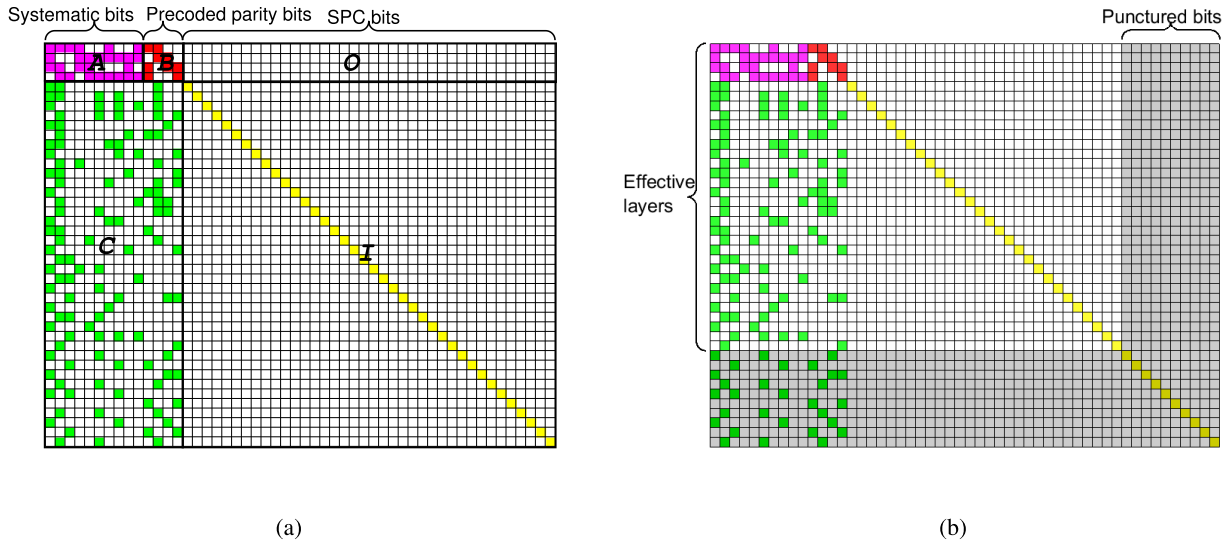


FIGURE 1. (a) BG2 for 5G NR LDPC codes and (b) Pruned BG2. White and colored squares represent all-zero square matrices and circularly-shifted identity matrices, respectively. The punctured bits in (b) result in the removal of shaded columns and rows.

Algorithm 1 Layered Decoding

Input : H and $\{\lambda_i : i \in \mathbb{Z}_{N-1}\}$.
Output: Estimate $\hat{v}$ and notification ξ
 // AP-LLR initialization
 1 $\gamma_i = \lambda_i$ **for** $i \in \mathbb{Z}_{N-1}$
 // C2V LLR initialization
 2 $\beta_{j \rightarrow i} = 0$ **for** $(j, i) \in \mathbb{Z}_{N-K-1} \times \mathcal{V}_j$
 3 **for** $t = 1, \dots, I_{\max}$ **do**
 4 **for** $l = \pi(\{0, 1, \dots, n-k-1\})$ **do**
 5 **for** $j \in \phi_l$ **do**
 6 $\{i_0, \dots, i_{d-1}\} \triangleq \mathcal{V}_j$ with $d = |\mathcal{V}_j|$
 7 // V2C LLR update
 8 $\alpha_{i_a \rightarrow j} = \gamma_{i_a} - \beta_{j \rightarrow i_a}$ **for** $a = 0, \dots, d-1$
 9 $s = \prod_{a=0}^{d-1} \text{sgn}(\alpha_{i_a \rightarrow j})$
 10 // C2V LLR update
 11 $\{\beta_{j \rightarrow i_a}, \gamma_{i_a} : a \in \mathbb{Z}_{d-1}\} = \text{SPA or MSA}(\{\alpha_{i_a \rightarrow j} : a \in \mathbb{Z}_{d-1}\}, s)$
 12 // Hard decisions
 13 $\hat{v}_i = \frac{1}{2}(1 - \text{sgn}(\gamma_i))$ **for** $i \in \mathbb{Z}_{N-1}$
 14 // Syndrome check
 15 **if** $H \times \hat{v}^T = \mathbf{0}$ **then**
 16 **return** $\xi = 1$ (decoding success) and $\hat{v}$
 17 **return** $\xi = 0$ (decoding failure)

Algorithm 2 SPA (With Forward-Backward Schedule)

Input : V2C LLRs $\{\alpha_{i_a \rightarrow j} : a \in \mathbb{Z}_{d-1}\}$ and the product s of their signs
Output: C2V LLRs $\{\beta_{j \rightarrow i_a} : a \in \mathbb{Z}_{d-1}\}$ and AP-LLRs $\{\gamma_{i_a} : a \in \mathbb{Z}_{d-1}\}$
 // C2V LLR update: Forward stage
 1 $R_1 = |\alpha_{i_0 \rightarrow j}|$
 2 **for** $a = 2, \dots, d-1$ **do**
 3 $R_a = f(|\alpha_{i_{a-1} \rightarrow j}|, R_{a-1})$
 4 // C2V LLR update: Backward stage
 5 $L_{d-2} = |\alpha_{i_{d-1} \rightarrow j}|$
 6 **for** $a = d-3, \dots, 0$ **do**
 7 $L_a = f(|\alpha_{i_{a+1} \rightarrow j}|, L_{a+1})$
 8 // C2V LLR update: Final stage
 9 $\beta_{j \rightarrow i_0} = s \cdot \text{sgn}(\alpha_{i_0 \rightarrow j}) \cdot L_0$
 10 $\beta_{j \rightarrow i_{d-1}} = s \cdot \text{sgn}(\alpha_{i_{d-1} \rightarrow j}) \cdot R_{d-1}$
 11 **for** $a = 1, \dots, d-2$ **do**
 12 $\beta_{j \rightarrow i_a} = s \cdot \text{sgn}(\alpha_{i_a \rightarrow j}) \cdot f(R_a, L_a)$
 13 // AP-LLR update
 14 $\gamma_{i_a} = \alpha_{i_a \rightarrow j} + \beta_{j \rightarrow i_a}$ **for** $a = 0, \dots, d-1$
 15 **return** $\{\beta_{j \rightarrow i_a}, \gamma_{i_a} : a \in \mathbb{Z}_{d-1}\}$

The operation $f(x_1, x_2)$ for SPA can be formulated as:

$$f(x_1, x_2) = 2 \times \tanh^{-1} \left(\tanh \frac{x_1}{2} \times \tanh \frac{x_2}{2} \right). \quad (1)$$

which involves complex hyperbolic tangent operations. The complexity of SPA depends on the number of calculations of $f(x_1, x_2)$.

2) MIN-SUM ALGORITHM

The CN finds the minimum value and the second minimum value from the received V2C LLR magnitudes and records

1) SPA WITH FORWARD-BACKWARD SCHEDULE

In Algorithm 2, two types of intermediate LLRs [18], denoted R and L , are computed in the forward stage (see Steps 1–3) and the backward stage (see Steps 4–6), respectively. In the final stage (see Steps 7–10), intermediate LLRs are combined to generate the C2V LLRs.

Algorithm 3 MSA

Input : V2C LLRs $\{\alpha_{i_a \rightarrow j} : a \in \mathbb{Z}_{d-1}\}$ and the product s of their signs

Output: C2V LLRs $\{\beta_{j \rightarrow i_a} : a \in \mathbb{Z}_{d-1}\}$ and AP-LLRs $\{\gamma_{i_a} : a \in \mathbb{Z}_{d-1}\}$

// Find the minimum value and its index

1 $m_1 = \min_{a \in \mathbb{Z}_{d-1}} \{|\alpha_{i_a \rightarrow j}|\}$

2 $a^* = \arg \min_{a \in \mathbb{Z}_{d-1}} \{|\alpha_{i_a \rightarrow j}|\}$

// Find the second minimum value

3 $m_2 = \min_{a \in \mathbb{Z}_{d-1} \setminus \{a^*\}} \{|\alpha_{i_a \rightarrow j}|\}$

// C2V LLR update

4 **for** $a = 0, \dots, d-1$ **do**

5 **if** $a = a^*$ **then**

6 $\beta_{j \rightarrow i_a} = s \cdot \text{sgn}(\alpha_{i_a \rightarrow j}) \cdot m_2$

7 **else**

8 $\beta_{j \rightarrow i_a} = s \cdot \text{sgn}(\alpha_{i_a \rightarrow j}) \cdot m_1$

// AP-LLR update

9 $\gamma_{i_a} = \alpha_{i_a \rightarrow j} + \beta_{j \rightarrow i_a}$ **for** $a = 0, \dots, d-1$

10 **return** $\{\beta_{j \rightarrow i_a}, \gamma_{i_a} : a \in \mathbb{Z}_{d-1}\}$

the VN providing the minimum value. Then, the CN passes back the second minimum value to the VN providing the minimum value, and the minimum value to the rest of the neighboring VNs. The pseudocode is presented in Algorithm 3.

III. REDUCED-COMPLEXITY LAYERED DECODING**A. DEACTIVATING VARIABLE NODES**

As illustrated by Step 10 of Algorithm 1, hard decisions are made based on AP-LLRs. A bit is decoded correctly with a high probability if the absolute value of its AP-LLR is large enough. For convenience, when referring to a “large” LLR, we are implying that the absolute value of the LLR is significant.

With the decoding iterations, AP-LLRs increase and eventually converge, indicating their stability in the subsequent iterations and resulting in stable decisions. Therefore, when an AP-LLR is large enough, it can be considered to deactivate the corresponding VN by skipping the relevant LLR updates subsequently, to reduce the decoding complexity. AP-LLRs converge at different rates across individual VNs, and some of them become large in just a few iterations while others stay small. Monitoring their convergence and deactivating VNs individually in each iteration causes additional complexity. In this paper, we construct the set of dVNs solely based on initial LLRs. The construction is completed before the decoding iterations start and remains unchanged during the decoding iterations.

1) BIT-LEVEL CHANNELS OF M -ary INPUT CHANNEL

We can treat the M -ary input channel, whose output is denoted by y , as a set of $\log_2(M)$ virtual binary-input

channels. The virtual channel $i \in \mathbb{Z}_{\log_2(M)-1}$ takes the bit $v^{(i)}$ as input and produces LLR $\lambda^{(i)}$ calculated as follows:²

$$\lambda^{(i)} = \log \frac{p(y|v^{(i)} = 0)}{p(y|v^{(i)} = 1)}. \quad (2)$$

Let $p(y|x)$ denote the transition probability of the M -ary input channel. For AWGN channels with the noise power spectral density N_0 and complex input symbols, the conditional probability used to calculate LLRs is given by

$$\begin{aligned} p(y|v^{(i)} = v) &= \frac{2}{M} \times \sum_{x \in \mathcal{X}(v^{(i)} = v)} p(y|x) \\ &= \frac{2}{M} \times \sum_{x \in \mathcal{X}(v^{(i)} = v)} \frac{1}{\pi N_0} e^{-\frac{|y-x|^2}{N_0}}, \end{aligned} \quad (3)$$

where v belongs to $\{0, 1\}$ and $\mathcal{X}(v^{(i)} = v)$ denotes the subset of the signal set in which $v^{(i)}$ is equal to v . We refer to these virtual channels as bit-level channels, and the bit position i as the level i .

The resulting bit-level channels demonstrate various levels of reliability, by checking the distributions of the corresponding LLRs. For instance, Fig. 2a shows that the bits assigned to the levels $\{0, 2\}$ in QAM-16 modulation observe more reliable bit-level channels than these assigned to the levels $\{1, 3\}$ due to their relatively larger LLR values. This phenomenon is amplified as the modulation order increases. As depicted in Fig. 2b, the difference in LLR values between the levels $\{0, 1\}$ and the levels $\{6, 7\}$ is significant in high-order modulations.

2) LAYERED DECODING WITH dVNs

Based on the aforementioned observation that AP-LLRs initialized by the LLRs from the most reliable bit-level channels tend to be large and imply correct decisions, we choose to deactivate the VNs allocated to these bit-level channels. However, there still exist small initial LLRs on the most reliable bit-level channels (see the magnified sub-figure of Fig. 2b). Therefore, among the VNs allocated to the most reliable bit-level channels, we deactivate these with initial LLRs exceeding a threshold.

Let ω denote the threshold and $\mathcal{L}^*$ denote the index set of the most reliable bit-level channels, e.g., $\mathcal{L}^* = \{0, 2\}$ and $\{0, 1\}$ in Figs. 2a and 2b, respectively. The index set of dVNs is obtained as

$$\mathcal{D} = \{i : (i \bmod \log_2(M)) \in \mathcal{L}^* \text{ and } |\lambda_i| > \omega\}.$$

The layered decoding with the set $\mathcal{D}$ of dVNs is described in Algorithm 4, where the pseudocode different from the (standard) layered decoding is highlighted in red. Specifically, Step 6 excludes dVNs from updating their C2V LLRs and AP-LLRs. Since both $\mathcal{V}_j$ and $\mathcal{D}$ are fixed, $\mathcal{V}_j \setminus \mathcal{D}$

²The utilized conditional probabilities are dedicated to BICM [19] which is the focus of this paper. Other coded modulation schemes employ distinct ones. For example, in the multilevel coded modulation [20], $p(y, v^{(0)}, \dots, v^{(i-1)} | v^{(i)} = 0 \text{ or } 1)$ is utilized.

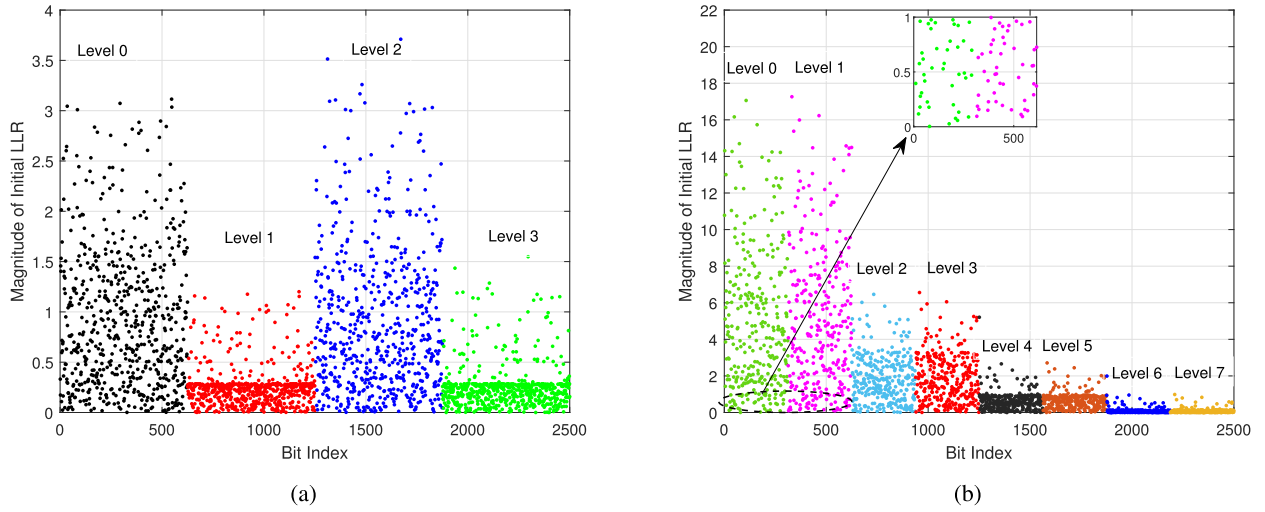


FIGURE 2. LLRs of bit-levels channel for (a) QAM-16 with Gray labeling and (b) QAM-256 with 5G NR labeling [2]. 2500 randomly generated bits are modulated and transmitted through AWGN channel of $E_s/N_0 = -4.5$ dB for (a) and 7.5 dB for (b). LLRs are calculated based on Equation (2).

Algorithm 4 Layered Decoding With dVNs

Input : H , $\{\lambda_i : i \in \mathbb{Z}_{N-1}\}$ and the set $\mathcal{D}$ of dVNs.

Output: Estimate $\hat{\mathbf{v}}$ and notification ξ

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1  $\gamma_i = \lambda_i$  for  $i \in [0 : N - 1]$ 
2  $\beta_{j \rightarrow i} = 0$  for  $(j, i) \in \mathbb{Z}_{N-K-1} \times \mathcal{V}_j$ 
3 for  $t = 1, 2, \dots, I_{\max}$  do
4   for  $l = \pi(\{0, 1, \dots, n - k - 1\})$  do
5     for  $j \in \phi_l$  do
6        $\{i_0, \dots, i_{d-1}\} \triangleq \mathcal{V}_j \setminus \mathcal{D}$  with  $d = |\mathcal{V}_j \setminus \mathcal{D}|$ 
7        $\alpha_{i_a \rightarrow j} = \gamma_{i_a} - \beta_{j \rightarrow i_a}$  for  $a = 0, \dots, d - 1$ 
8        $s = \prod_{a=0}^{d-1} \text{sgn}(\alpha_{i_a \rightarrow j}) \cdot \prod_{i \in \mathcal{V}_j \cap \mathcal{D}} \text{sgn}(\gamma_i)$ 
9        $\{\beta_{j \rightarrow i_a}, \gamma_{i_a} : a \in \mathbb{Z}_{d-1}\} = \text{SPA or}$ 
         $\text{MSA}(\{\alpha_{i_a \rightarrow j} : a \in \mathbb{Z}_{d-1}\}, s)$ 
10     $\hat{v}_i = \frac{1}{2}(1 - \text{sgn}(\gamma_i))$  for  $i \in \mathbb{Z}_{N-1}$ 
11    if  $H \times \hat{\mathbf{v}}^T = \mathbf{0}$  then
12      return  $\xi = 1$  (decoding success) and  $\hat{\mathbf{v}}$ 
13 return  $\xi = 0$  (decoding failure)
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can be calculated and stored as a lookup table ahead of the decoding iterations. We incorporate this operation within the iteration to facilitate the description of the proposed method. The signs of the V2C LLRs from dVNs are still needed, as indicated by $\prod_{i \in \mathcal{V}_j \cap \mathcal{D}} \text{sgn}(\gamma_i)$ at Step 8. At Step 9, when

SPA is applied, the calculations of $f(x_1, x_2)$ related to dVNs are omitted. When MSA is applied, the searching space for locating the minimum value and the second minimum value is narrowed.

3) THRESHOLD DESIGN

By adjusting the threshold, we can manage the trade-off between the complexity and the performance of the

layered decoding. Lowering the threshold reduces the decoding complexity by increasing the number of dVNs. On the other hand, it trades off the decoding performance due to no further AP-LLR updates for the corresponding dVNs. In the sequel, we design thresholds to optimize the complexity-performance trade-off.

The design relies on the characteristics of the bit-level channels derived in Section III-A1. The PDF of the initial LLR $\lambda^{(i)}$, when transmitting bit v through the bit-level channel i , is utilized and given by:

$$p(\lambda^{(i)} | v^{(i)} = v) = \int_{\mathbb{C}} p(y | v^{(i)} = v) \times \mathbb{1}\left(\lambda^{(i)} = \log\left(\frac{p(y | v^{(i)} = 0)}{p(y | v^{(i)} = 1)}\right)\right) dy, \quad (4)$$

where $\mathbb{C}$ represents the set of complex numbers, and $\mathbb{1}(\cdot)$ equals to 1 when the argument is true and 0 otherwise.

Given a threshold ω , the probability of deactivating a VN assigned to the most reliable bit-level channel $i \in \mathcal{L}^*$ is given by

$$P^{(i)} = \frac{1}{2} \times \sum_{v=0}^1 \left[\int_{|\lambda^{(i)}| > \omega} p(\lambda^{(i)} | v^{(i)} = v) d\lambda^{(i)} \right]. \quad (5)$$

In BICM systems with M -ary modulation, N codeword bits are split into $\frac{N}{\log_2(M)}$ bit streams, each for a bit-level channel. Without loss of generality, N is assumed to be divisible by $\log_2(M)$.³ Based on Equation (5), the average number of dVNs is obtained as follows:

$$D = \frac{N}{\log_2(M)} \times \sum_{i \in \mathcal{L}^*} P^{(i)}. \quad (6)$$

³In situations where N is not divisible by $\log_2(M)$, zero bits can be attached to the end of codeword to ensure complete mapping.

For a dVN, the probability of making an incorrect decision averaged over the levels $\mathcal{L}^*$ is expressed as follows:

$$P_e = \frac{1}{|\mathcal{L}^*|} \times \sum_{i \in \mathcal{L}^*} \left[\int_{-\infty}^{-\omega} p(\lambda^{(i)} | v^{(i)} = 0) d\lambda^{(i)} + \int_{\omega}^{\infty} p(\lambda^{(i)} | v^{(i)} = 1) d\lambda^{(i)} \right]. \quad (7)$$

The first and second integrals in the summation of Equation (7) correspond to the transmission of zero and non-zero bits, respectively. It is noted that VNs are not classified as dVNs when $-\omega < \lambda^{(i)} < 0$ in case of sending a zero bit and $0 < \lambda^{(i)} < \omega$ in case of sending a non-zero bit. The signs of their AP-LLRs will be corrected during the decoding iterations.

The threshold ω is determined so as to result in a desired error probability P_d , which represents the likelihood of at least one error occurring among D decisions to categorize some of VNs as dVNs. It is calculated as

$$P_d = 1 - (1 - P_e)^D. \quad (8)$$

By carefully selecting the design parameter P_d , we are able to reduce the complexity while maintaining a similar performance, as the results shown in Section V.

4) SHORTENED VARIABLE NODES

To accommodate rate and length flexibility, the shortening technique may be applied to LDPC codes as in the case of 5G NR LDPC codes described in Section II-B2. The shortening is realized by freezing certain systematic bits to zeros. Because the positions and the values of the shortened bits are revealed to the decoder, the corresponding initial LLRs are set to infinity. In the proposed method, we also characterize the shortened VNs as the set $\mathcal{D}$ of dVNs.

B. HYBRID SUM-PRODUCT AND MIN-SUM ALGORITHM

The primary advantage of SPA is its ability to achieve near-optimal performance by accurately computing the C2V LLRs. However, the involved hyperbolic tangent operations require a significant amount of decoding time and resources, which limits the decoding throughput. MSA, which approximates the hyperbolic tangent operations, can enhance the decoding throughput at the expense of the performance.

Based on the fact that each layer in the layered decoding are updated independently, our hSPMSA applies either SPA or MSA to each respective layer. The layers employing SPA and MSA are referred to as the SPA and MSA layers, respectively. By carefully determining the SPA and MSA layers, hSPMSA can exploit both the high computation accuracy offered by SPA and the reduced complexity offered by MSA. We use $\mathcal{H}_{SPA} \subseteq \mathbb{Z}_{n-k-1}$ and its complement $\mathcal{H}_{MSA} = \mathbb{Z}_{n-k-1} \setminus \mathcal{H}_{SPA}$ to denote the set of the SPA and MSA layers, respectively. Algorithm 5 describes pseudocode of the layered decoding with hSPMSA, highlighting the difference in red compared with Algorithm 1. At Step 13,

Algorithm 5 Layered Decoding With hSPMSA

Input : $H, \{\lambda_i : i \in \mathbb{Z}_{N-1}\}, \mathcal{H}_{SPA}, \mathcal{H}_{MSA}$.
Output: Estimate $\hat{\mathbf{v}}$ and notification ξ

```

1  $\gamma_i = \lambda_i$  for  $i \in \mathbb{Z}_{N-1}$ 
2  $\beta_{j \rightarrow i} = 0$  for  $(j, i) \in \mathbb{Z}_{N-K-1} \times \mathcal{V}_j$ 
3 for  $t = 1, \dots, I_{\max}$  do
4   for  $l = \pi(\{0, 1, \dots, n-k-1\})$  do
5     if  $l \in \mathcal{H}_{SPA}$  then
6       UpdateLLR = SPA
7     if  $l \in \mathcal{H}_{MSA}$  then
8       UpdateLLR = MSA
9     for  $j \in \phi_l$  do
10       $\{i_0, \dots, i_{d-1}\} \triangleq \mathcal{V}_j$  with  $d = |\mathcal{V}_j|$ 
11       $\alpha_{i_a \rightarrow j} = \gamma_{i_a} - \beta_{j \rightarrow i_a}$  for  $a = 0, \dots, d-1$ 
12       $s = \prod_{a=0}^{d-1} \text{sgn}(\alpha_{i_a \rightarrow j})$ 
13       $\{\beta_{j \rightarrow i_a}, \gamma_{i_a} : a \in \mathbb{Z}_{d-1}\} =$ 
        UpdateLLR( $\{\alpha_{i_a \rightarrow j} : a \in \mathbb{Z}_{d-1}\}, s$ )
14    $\hat{v}_i = \frac{1}{2}(1 - \text{sgn}(\gamma_i))$  for  $i \in \mathbb{Z}_{N-1}$ 
15   if  $H \times \hat{\mathbf{v}}^T = \mathbf{0}$  then
16     return  $\xi = 1$  (decoding success) and  $\hat{\mathbf{v}}$ 
17 return  $\xi = 0$  (decoding failure)
```

the rules of updating LLRs for a layer follow SPA if the layer is a SPA layer (see Step 6), and follow MSA otherwise (see Step 8).

We calculate the overall computational complexity of hSPMSA as a function of degrees of the layers and the number of SPA and MSA layers. By analyzing the individual impacts of the number of the SPA layers and the summation of the degrees of the SPA layers on the complexity, we develop and present a degree-based method for determining the SPA layers and the MSA layers.

Fig. 3 illustrate the three stages of SPA with forward-backward schedule for a degree- d CN. Notably, if either input of $f(x_1, x_2)$ is infinite, the output of the function is equal to the other input. Consequently, two calculations of $f(x_1, x_2)$ can be saved in each stage: $R_0 = +\infty$ and $R_1 = f(\alpha_0, +\infty) = \alpha_0$ in the forward stage; $L_{d-1} = +\infty$ and $L_{d-2} = f(+\infty, \alpha_{d-1}) = \alpha_{d-1}$ in the backward stage; $\beta_0 = f(L_0, +\infty) = L_0$ and $\beta_{d-1} = f(+\infty, R_{d-1}) = R_{d-1}$ in the final stage. This results in $(d-2)$ calculations of $f(x_1, x_2)$ for each of the three stages.

When applying MSA to a degree- d CN, $d-1$ comparisons and $d-2$ comparisons are needed to find the minimum value and the second minimum value, respectively.

For QC-LDPC codes with the lifting size Z , the number of operations performed by hSPMSA per iteration is formulated as:

$$C \cdot Z \cdot \sum_{l \in \mathcal{H}_{SPA}} 3(d_l - 2) + Z \cdot \sum_{l \in \mathcal{H}_{MSA}} (2d_l - 3), \quad (9)$$

where C is the number of FLOPs needed to compute $f(x_1, x_2)$ and d_l represents the degree of the layer l . The first item

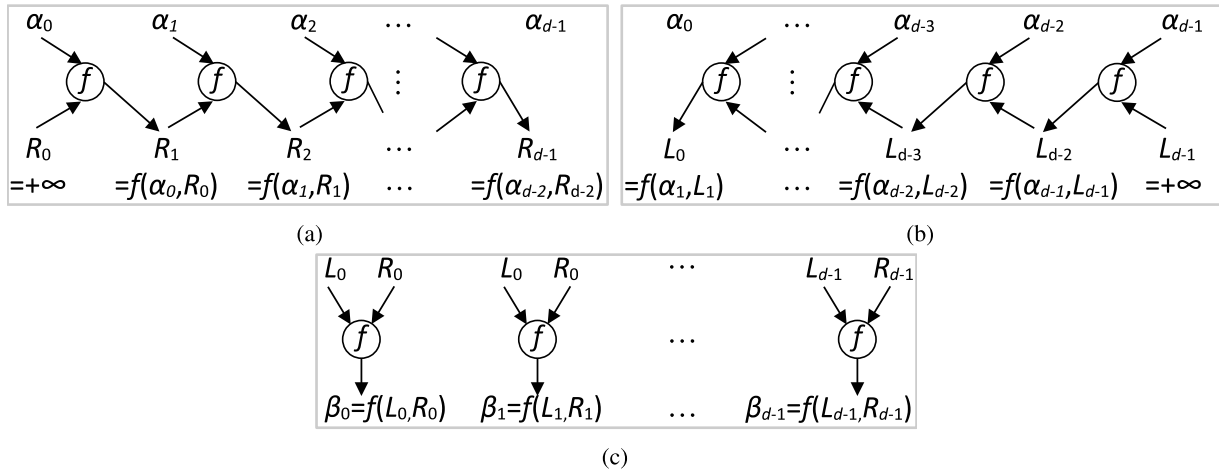


FIGURE 3. (a) Forward stage, (b) backward stage and (c) final stage for SPA with forward-backward schedule. $\{\alpha_0, \dots, \alpha_{d-1}\}$ denote the V2C LLRs for a degree- d CN. The intermediate LLRs $\{R_1, \dots, R_{d-1}\}$ ($\{L_0, \dots, L_{d-1}\}$) are calculated recursively from the left (right) to the right (left) in the forward (backward) stage with the initialization $R_0 = L_{d-1} = +\infty$. In the final stage, the C2V LLRs $\{\beta_0, \dots, \beta_{d-1}\}$ are produced.

and the second item in (9) represent the complexity caused by SPA and MSA, respectively. Due to a large value of C (specified in Section IV), the SPA layers predominantly contribute to the complexity of hSPMSA. In the sequel, we focus on analyzing the first item in (9) which can be reformulated as follows:

$$C \cdot Z \cdot \sum_{l \in \mathcal{H}_{\text{SPA}}} 3(d_l - 2) = C \cdot Z \cdot (3 \sum_{l \in \mathcal{H}_{\text{SPA}}} d_l - 6|\mathcal{H}_{\text{SPA}}|) \\ = C \cdot Z \cdot (3n_{\text{SPA}} - 6|\mathcal{H}_{\text{SPA}}|). \quad (10)$$

where n_{SPA} represents the summation of degrees across all SPA layers and $-6|\mathcal{H}_{\text{SPA}}|$ is due to the omitted hyperbolic tangent calculations mentioned above.

This paper aims to minimize the complexity of hSPMSA conditioned on a desired performance. The error-correcting capability of hSPMSA depends on n_{SPA} , as evidenced by the numerical findings presented in Section V-B. It is observed that various constructions of $\mathcal{H}_{\text{SPA}}$ and $\mathcal{H}_{\text{MSA}}$, resulting from the same value of n_{SPA} , yield a very similar performance.

According to Equation 10, when setting n_{SPA} as a specific value to maintain a desired performance, maximizing the number of SPA layers, i.e., $|\mathcal{H}_{\text{SPA}}|$, saves most calculations of $f(x_1, x_2)$. We accomplish this by prioritizing the application of SPA to lower-degree layers. The degree-based method is detailed in Algorithm 6. At Step 2, if multiple layers have the same degree, we arrange them based on the processing order $\pi(\{0, 1, \dots, n-k-1\})$ of layers.

C. JOINTLY REDUCING COMPLEXITY

The two proposed methods, dVN and hSPMSA, can be employed together to achieve combined complexity reduction without introducing any conflicts, since their designs are VN-oriented and CN-oriented, respectively. The joint method is described in Algorithm 7.

Algorithm 6 Design of hSPMSA

Input : B and n_{SPA}
Output: $\mathcal{H}_{\text{SPA}}$ and $\mathcal{H}_{\text{MSA}}$
 // Compute the degrees of layers
 1 $d_l = |\{B_{l,i} \geq 0 : i \in \mathbb{Z}_{n-1}\}|$ for $l = 0, \dots, n-k-1$
 // Sort the layers by degree
 2 $\{l_{(0)}, \dots, l_{(n-k-1)}\} = \arg \text{sort}\{d_0, \dots, d_{n-k-1}\}$
 // Initialize $\mathcal{H}_{\text{SPA}}$
 3 $\mathcal{H}_{\text{SPA}} = \{\}$
 // Count the cumulative degrees involved in SPA
 4 $n_{\text{count}} = 0$
 // Add low-degree layers into $\mathcal{H}_{\text{SPA}}$ until n_{SPA} is met
 5 **for** $l = l_{(0)}, l_{(1)}, \dots, l_{(n-k-1)}$ **do**
 6 $\mathcal{H}_{\text{SPA}} = \mathcal{H}_{\text{SPA}} \cup l$
 7 $n_{\text{count}} = n_{\text{count}} + d_l$
 8 **if** $n_{\text{count}} \geq n_{\text{SPA}}$ **then**
 9 **break**
 // Construct $\mathcal{H}_{\text{MSA}}$
 10 $\mathcal{H}_{\text{MSA}} = \mathbb{Z}_{n-k-1} \setminus \mathcal{H}_{\text{SPA}}$
 11 **return** $\mathcal{H}_{\text{SPA}}$ and $\mathcal{H}_{\text{MSA}}$

IV. COMPLEXITY ANALYSIS

The number of FLOPs serves as the metric to assess the decoding complexity. A prevalent method to implement Equation (1) for SPA is reformulating it as follows:

$$f(x_1, x_2) = \min(x_1, x_2) - \log \left(\frac{1 + e^{-(x_1 + x_2)}}{1 + e^{-|x_1 - x_2|}} \right),$$

which requires $C = 12$ operations including one comparison, seven additions and subtractions, two exponentiations, one logarithm, and one division. For QC-LDPC codes with the lifting size Z , processing a SPA layer of degree d_l requires

Algorithm 7 Layered Decoding With hSPMSA and dVN

Input : $H, \{\lambda_i : i \in \mathbb{Z}_{N-1}\}, \mathcal{H}_{SPA}, \mathcal{H}_{MSA}, \mathcal{D}$.
Output: $\hat{v}, \xi$

```

1  $\gamma_i = \lambda_i$  for  $i \in \mathbb{Z}_{N-1}$ 
2  $\beta_{j \rightarrow i} = 0$  for  $(j, i) \in \mathbb{Z}_{N-K-1} \times \mathcal{V}_j$ 
3 for  $t = 1, \dots, I_{\max}$  do
4   for  $l = \pi(\{0, 1, \dots, n-k-1\})$  do
5     if  $l \in \mathcal{H}_{SPA}$  then
6       UpdateLLR = SPA
7     if  $l \in \mathcal{H}_{MSA}$  then
8       UpdateLLR = MSA
9     for  $j \in \phi_l$  do
10       $\{i_0, \dots, i_{d-1}\} \triangleq \mathcal{V}_j \setminus \mathcal{D}$  with  $d = |\mathcal{V}_j \setminus \mathcal{D}|$ 
11       $\alpha_{i_a \rightarrow j} = \gamma_{i_a} - \beta_{j \rightarrow i_a}$  for  $a = 0, \dots, d-1$ 
12       $s = \prod_{a=0}^{d-1} \text{sgn}(\alpha_{i_a \rightarrow j}) \cdot \prod_{i \in \mathcal{V}_j \cap \mathcal{D}} \text{sgn}(\gamma_i)$ 
13       $\{\beta_{j \rightarrow i_a}, \gamma_{i_a} : a \in \mathbb{Z}_{d-1}\} =$ 
        UpdateLLR( $\{\alpha_{i_a \rightarrow j} : a \in \mathbb{Z}_{d-1}\}, s$ )
14     $\hat{v}_i = \frac{1}{2}(1 - \text{sgn}(\gamma_i))$  for  $i \in \mathbb{Z}_{N-1}$ 
15    if  $H \times \hat{v}^T = \mathbf{0}$  then
16      return  $\xi = 1$  (decoding success) and  $\hat{v}$ 
17 return  $\xi = 0$  (decoding failure)
```

TABLE 1. FLOP list.

Algorithms	The required number of FLOPs
SPA	$\bar{t} \sum_{l=0}^{n-k-1} 36Z(d_l - 2)$
SPA incorporating dVNs	$\bar{t} \sum_{l=0}^{n-k-1} \left(36 \sum_{j \in \phi_l} (\mathcal{V}_j \setminus \mathcal{D} - 2) \right)$
hSPMSA	$\bar{t} \left(\sum_{l \in \mathcal{H}_{SPA}} 36Z(d_l - 2) + \sum_{l \in \mathcal{H}_{MSA}} Z(2d_l - 3) \right)$
hSPMSA incorporating dVNs	$\bar{t} \left[\sum_{l \in \mathcal{H}_{SPA}} \left(36 \sum_{j \in \phi_l} (\mathcal{V}_j \setminus \mathcal{D} - 2) \right) + \sum_{l \in \mathcal{H}_{MSA}} \left(\sum_{j \in \phi_l} (2 \mathcal{V}_j \setminus \mathcal{D} - 3) \right) \right]$

$Z \times 12 \times 3 \times (d_l - 2)$ FLOPs. Meanwhile, $Z \times (2d_l - 3)$ comparisons are needed to find the minimum value and the second minimum value, when processing a MSA layer.

Given the set $\mathcal{D}$ of dVNs, only the set $\mathcal{V}_j \setminus \mathcal{D}$ of VNs, which are active and connected to the CN j , are involved in the processing of the CN j . Therefore, $f(x_1, x_2)$ is calculated by $|\mathcal{V}_j \setminus \mathcal{D}| - 2$ times in each stage of the forward-backward schedule for SPA, and the number of comparisons used to find two minima is reduced to $2 \times |\mathcal{V}_j \setminus \mathcal{D}| - 3$ for MSA. The number

of FLOPs required to process a SPA layer and a MSA layer turn into $12 \times \sum_{j \in \phi_l} 3 \times (|\mathcal{V}_j \setminus \mathcal{D}| - 2)$ and $\sum_{j \in \phi_l} (2 \times |\mathcal{V}_j \setminus \mathcal{D}| - 3)$, respectively.

The number of iterations also contribute to the decoding complexity. The syndrome check can halt the decoding procedures prior to reaching the maximum number of iterations, causing the value of t in Algorithms 1, 4, 5 and 7 to vary across different SNRs, channel realizations and decoding algorithms. Therefore, we consider the average number of iterations (ANI) $\bar{t}$, which can be calculated empirically. The overall complexity in terms of FLOPs for the layered decoding with SPA and the proposed methods are summarized in Table 1.

V. SIMULATIONS AND DISCUSSIONS

The encoding procedures for the evaluations of the proposed methods comply with the 5G NR standard [2], which incorporates CRC encoding, systematic puncturing,⁴ rate matching and rectangular interleaving. The proposed methods are applicable to any code rate. However, the decoding procedures for high-rate codes subtly impact the system throughput, since the number of effective layers is small, as outlined in Section II-B2. This paper focuses on evaluating 5G NR LDPC codes with low-to-moderate code rates $\{0.3, 0.4, 0.5\}$ across information sizes $\{200, 300, \dots, 1000\} \cup \{1200, 1400, \dots, 8400\}$. QPSK, QAM-64 and QAM-256 modulations, whose mapping rules follow the 5G NR standard, are employed.

During the layered decoding, the sequence $\pi(\{0, 1, \dots, n-k-1\})$ utilized to determine the order of updating layers refers to [21] Section III-B. The syndrome check is conducted by using the sub-matrix $[A, B]$ instead of the entire parity check matrix. Specifically, a decoding success is declared if $[A, B] \times \hat{v}^T = \mathbf{0}$ is satisfied. The maximum number of iterations is set as 10.

The FLOP analysis in Section IV is utilized to evaluate the complexity of the layered decoding. The number of FLOPs naturally grow with the information size. To illustrate the reduction in FLOPs across information sizes ranging from 200 to 8400 on a consistent scale, the *ratio* of the reduced FLOPs relative to a baseline is presented. To assess the impact of the proposed methods on the error-correcting performance, the SNRs required to attain a target BLER are also presented. Following the requirement of the 3GPP NR enhanced mobile broadband (eMBB) [22], we set the target BLER as 10^{-2} . ANI $\bar{t}$ is equal to the empirical mean of the number of iterations at the corresponding desired SNR. In every instance of MSA, the offset MSA with the offset value 0.5 is utilized.

A. DEACTIVATING VARIABLE NODES

Fig. 4 presents the comparison between SPA incorporating dVNs (SPA-dVNs) and SPA, under QAM-256 modulation.

⁴The bits corresponding to the first two columns of BG1 and BG2 are always punctured.

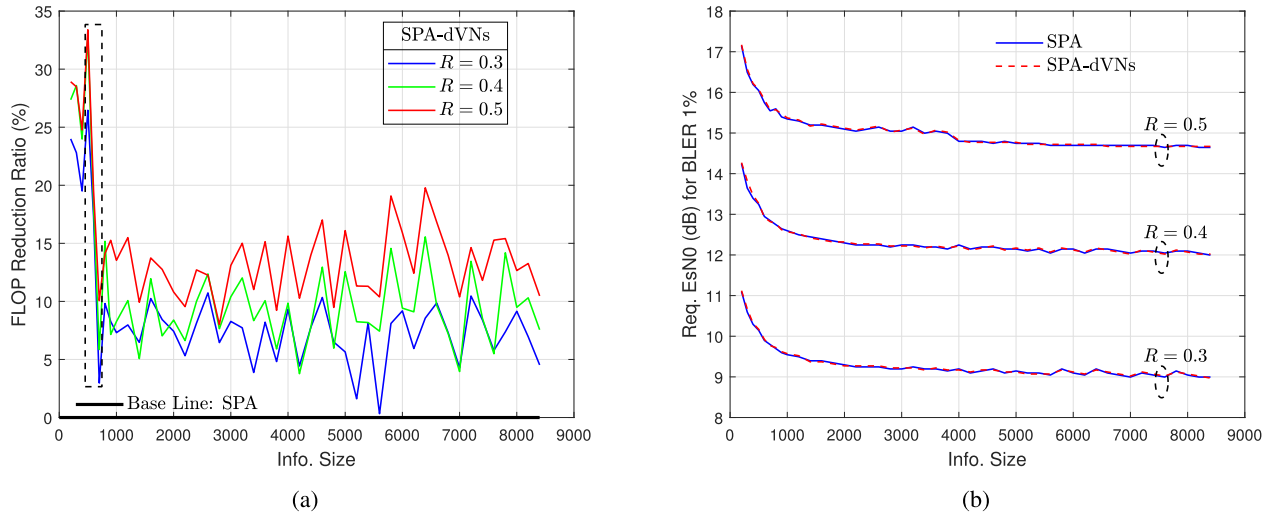


FIGURE 4. (a) Complexity and (b) performance comparison for QAM-256 between SPA-dVNs with $P_d = 10^{-3}$ and SPA.

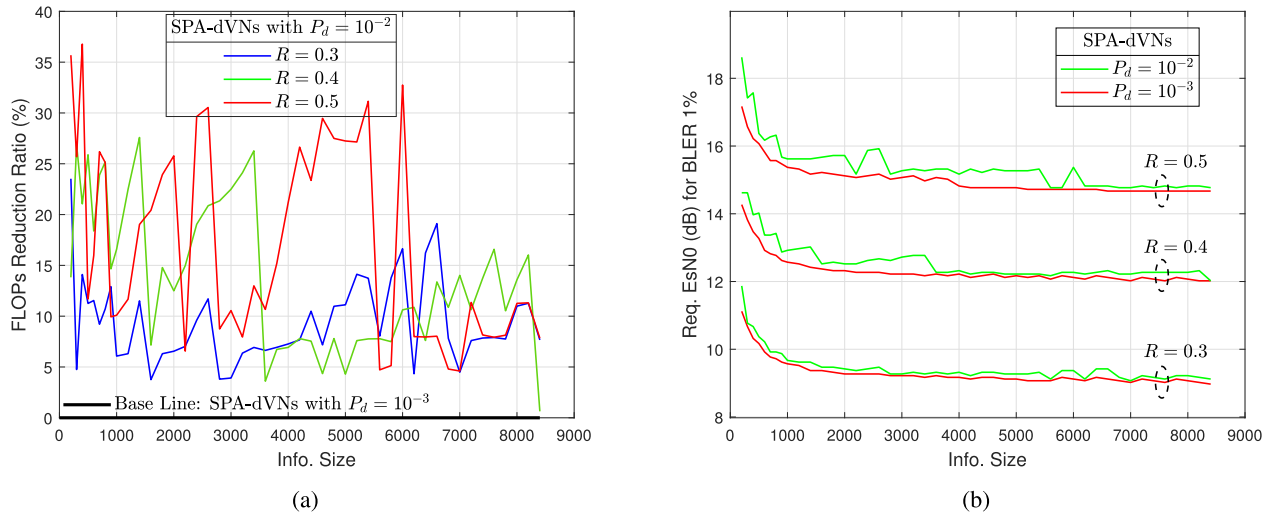


FIGURE 5. (a) Complexity and (b) performance comparison for SPA-dVNs with QAM-256 between $P_d = 10^{-2}$ and $P_d = 10^{-3}$.

The tuning parameter P_d for SPA-dVNs is set as 10^{-3} . The threshold for SPA-dVNs is optimized and adjusted during the evaluations based on Equation (8).

SPA-dVNs achieves up to 33% FLOP reduction over SPA as depicted in Fig. 4a, while maintaining nearly identical performance as shown in Fig. 4b. The FLOP reduction ratio generally grows with the code rate, since the number of effective layers decreases, reducing the total number of edges. The significant change of the number of effective layers also causes the drops of the FLOP reduction ratio, inside the dashed rectangle in Fig. 4a. For example, the number of effective layers increases from 18 to 25, when the information size changes from 500 to 700 at the code rate 0.3.

Relaxing P_d enables SPA-dVNs to further reduce the complexity, since the resulting LLR thresholds used to identify dVNs become smaller. As shown in Fig. 5a,

increasing the value of P_d from 10^{-3} to 10^{-2} leads to extra FLOP reduction for SPA-dVNs. However, SPA-dVNs configured with $P_d = 10^{-3}$ shows significant performance gains (up to 1.6 dB) over the configuration of $P_d = 10^{-2}$, as shown in Fig. 5b. There are two reasons for the gains. First, setting $P_d = 10^{-3}$ guarantees the initial LLRs of dVNs are large enough, allowing them to converge toward correct decisions. Second, the C2V LLR updates are dominated by the V2C LLRs of small values, indicating that dVNs delivering large LLRs have a trivial impact on the AP-LLR updates for the active VNs. When increasing the value of P_d to 10^{-2} , additional VNs with small initial LLRs are deactivated. As a result, incorrect decisions caused by small initial LLRs are introduced, due to the fact that dVNs are prohibited from receiving intrinsic LLRs to correct the signs of their AP-LLRs.

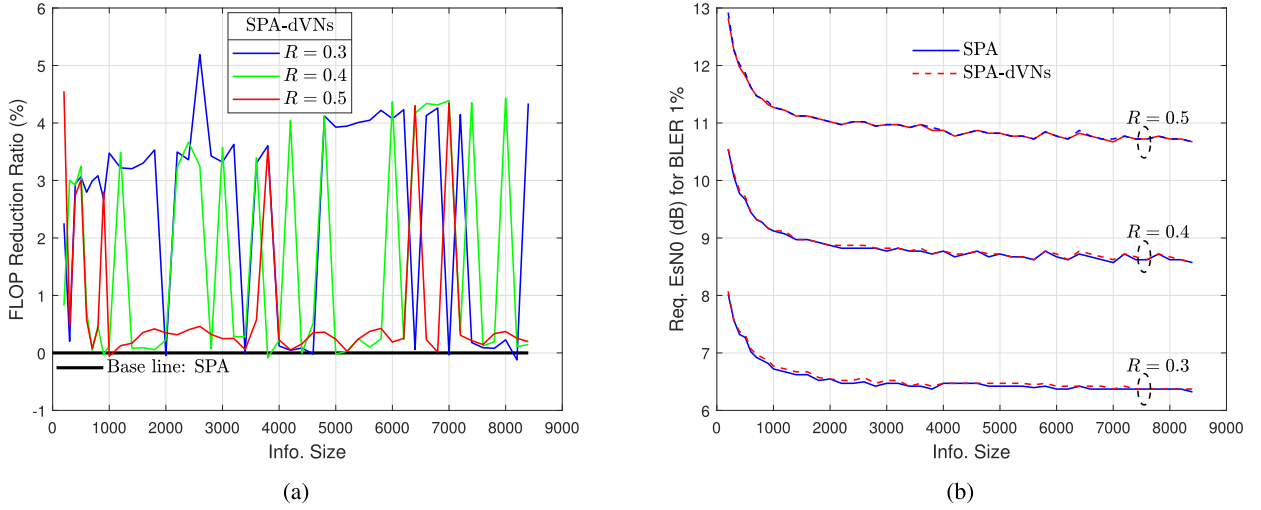


FIGURE 6. (a) Complexity and (b) performance comparison for QAM-64 between SPA-dVNs with $P_d = 10^{-3}$ and SPA.

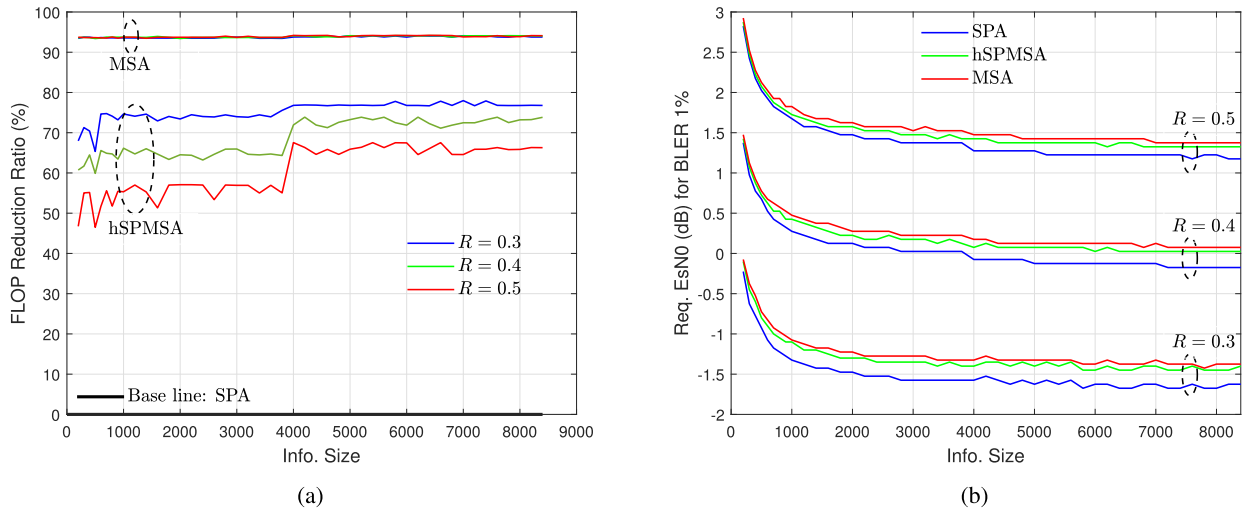


FIGURE 7. (a) Complexity and (b) performance comparison between hSPMSA, MSA and SPA for QPSK. In hSPMSA, $n_{SPA} = 76$ for BG1 and 36 for BG2.

The application of the proposed method is not exclusive to high-order modulations. For example, Fig. 6 illustrates the results for QAM-64 modulation. Compared with SPA, SPA-dVNs can reduce the number of FLOPs by up to 5.2% (see Fig. 6a), while keeping the performance degradation within 0.1 dB (see Fig. 6b). Fig. 6a shows a few cases where SPA-dVNs is marginally more complex than SPA. This is due to an increase in ANI at the desired SNR. For example, at the code rate 0.5 and the information size 8200, the values of ANI for SPA-dVNs and SPA are equal to 7.23 and 7.21, respectively. SPA-dVNs equipped with QAM-16 modulation did not produce promising results in our experiments.

B. HYBRID SUM-PRODUCT AND MIN-SUM ALGORITHM

Fig. 7 compares SPA, MSA and hSPMSA. We determine the SPA layers and the MSA layers for hSPMSA by using the

degree-based method proposed in Section III-B, where the design parameter n_{SPA} is set as 76 for BG1 and 36 for BG2. In Fig. 7a, hSPMSA exhibits a complexity level falling between SPA and MSA. So does the performance illustrated in Fig. 7b. The FLOP reduction ratio achieved by hSPMSA decreases with the code rate, due to the shrinking (growing) percentage of MSA (SPA) layers. For 5G NR LDPC codes with low-to-moderate code rates, the choice of BG1 and BG2 depends on the information size: If it is less than or equal to 3824, BG2 is used; Otherwise, BG1 can be used. This choice leads to the sudden rise of the FLOP reduction ratio at the information size 4000.

To show the flexibility of hSPMSA in managing the complexity-performance trade-off by adjusting n_{SPA} , in Fig. 8, we increase the value of n_{SPA} from 76 to 150 for BG1, and from 36 to 50 for BG2. According to the degree-based method, raising the value of n_{SPA} causes SPA

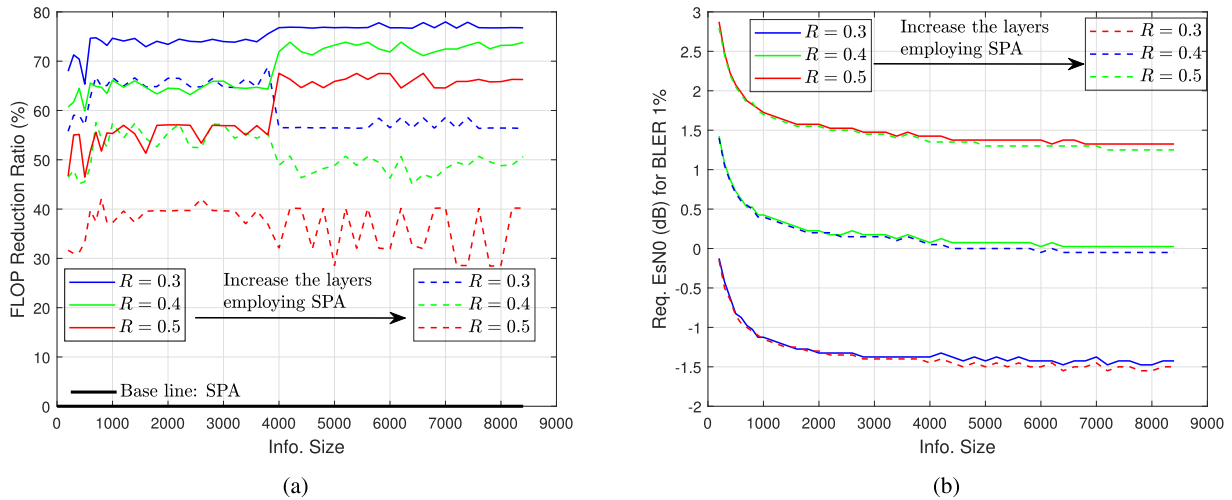


FIGURE 8. (a) Complexity and (b) performance comparison between hSPMSA and SPA for QPSK. In hSPMSA, n_{SPA} is increased from 76 to 150 for BG1, and from 36 to 50 for BG2.

to be employed by more layers, whereas MSA is applied to fewer layers. Therefore, hSPMSA shows an improved performance (see Fig. 8b) with an increased complexity (see Fig. 8a).

The complexity-performance trade-off offered by hSPMSA also relies on methods used to determine the SPA and the MSA layers. Fig. 9 compares the proposed degree-based method, labeled as “SPA for the low-degree layers”, with two intuitive methods, labeled as “SPA for the randomly-selected layers” and “SPA for the first four layers”. The three methods are implemented under the same constraint: $n_{\text{SPA}} = 76$ for BG1 and $n_{\text{SPA}} = 36$ for BG2. For “SPA for the randomly-selected layers”, SPA is applied to the layers randomly selected from the effective layers and MSA is applied to the remaining layers. In BG1 and BG2, the submatrices $\mathbf{A}$ and $\mathbf{B}$ (see Fig. 1a) serve as the *core* components in correcting errors. For “SPA for the first four layers”, SPA is applied to the layers containing the core components, i.e., the first four layers, and MSA is applied to the remaining layers. Because SPA provides more accurate LLR calculations than MSA, applying SPA to the core components may allow us to preserve the error-correcting ability of the layer decoding as much as possible. In Fig. 9b, the three methods perform closely, which indicates that the performance of the layered decoding relies on n_{SPA} , i.e., the total number of edges involved in SPA. The superiority of the degree-based method in reducing complexity is shown in Fig. 9a, where up to 20% FLOP reduction over the random-selection based method are achieved. Based on Equation 10, we infer that for a particular value of n_{SPA} , the larger the number of SPA layers, the lower the complexity. The degree-based method maximizes the number of SPA layers and hence achieves the lowest complexity, while applying SPA to the core components results in the highest complexity due to the smallest number of SPA layers.

Since the number of layers in base graphs of QC-LDPC codes is limited, a lookup table containing the indices of the

TABLE 2. SPA layers for hSPMSA.

k'	BG1	k'	BG2
22	4,7,10,12–14,16–21	9	2,4–8
23	4,7,10,13,14,16–22	10	4–9
24	4,7,10,13,14,16–23	11	4–10
32	4,13,16,19,20	12	4,5,7–11
	22,23,25–31	13	4,5,8–12
33	4,16,19–23,25–32	14	4,8–13
34	4,16,19,20,22	15	4,8–10,12–14
	23,25–33	16–18	4,8–13,15
35	4,16,19,22,23	19	4,8–10,12,13,15,18
	25–34	20	4,8,9,12,13,15,18,19
46	4,22,23,25–32	22	4,8,12,15,18–21
	35,37,40,42,45	24,25	4,8,12,15,18–20,22,23
		26	4,8,12,15,18–20,22,25

SPA and MSA layers can be created offline. As an example, Table 2 lists the SPA layers employed in our simulations. They are determined by using the degree-based method, where n_{SPA} is set as 76 for BG1 and 36 for BG2. In the table, k' denotes the number of effective layers. Given the index set $\mathcal{H}_{\text{SPA}}$ of SPA layers, the index set $\mathcal{H}_{\text{MSA}}$ of MSA layers is equal to $\mathbb{Z}_{k'} \setminus \mathcal{H}_{\text{SPA}}$.

C. JOINTLY REDUCING COMPLEXITY

Fig. 10 demonstrates that incorporating dVNs within hSPMSA (referred to as hSPMSA-dVNs) leads to an additional reduction in FLOPs by up to 26%, while inducing

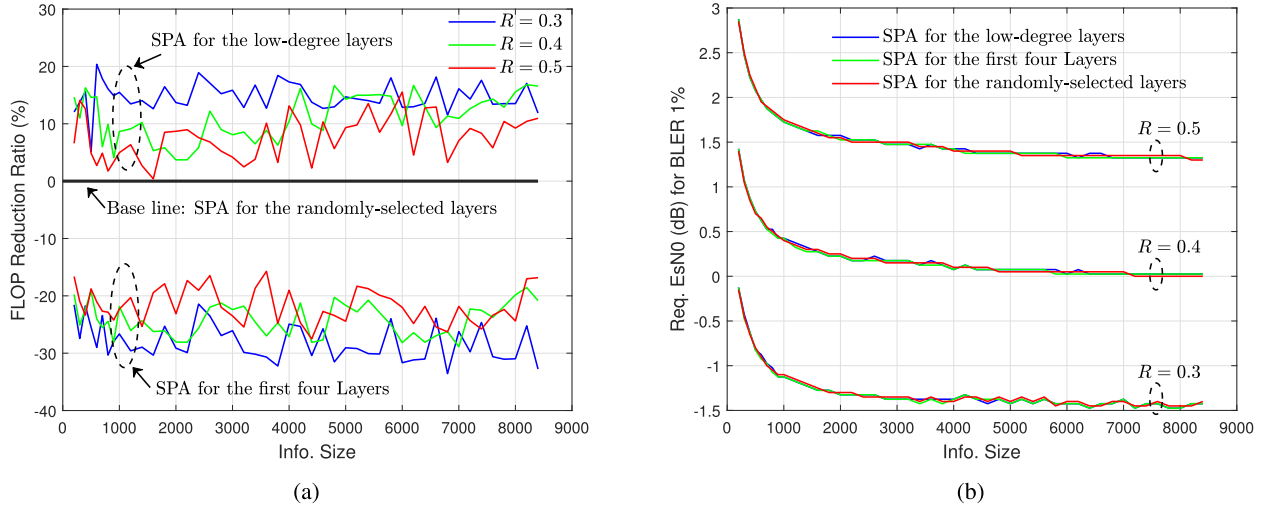


FIGURE 9. (a) Complexity and (b) performance comparison for QPSK between three distinct methods used to determine the SPA and the MSA layers for hSPMSA. $n_{\text{SPA}} = 76$ for BG1 and 36 for BG2.

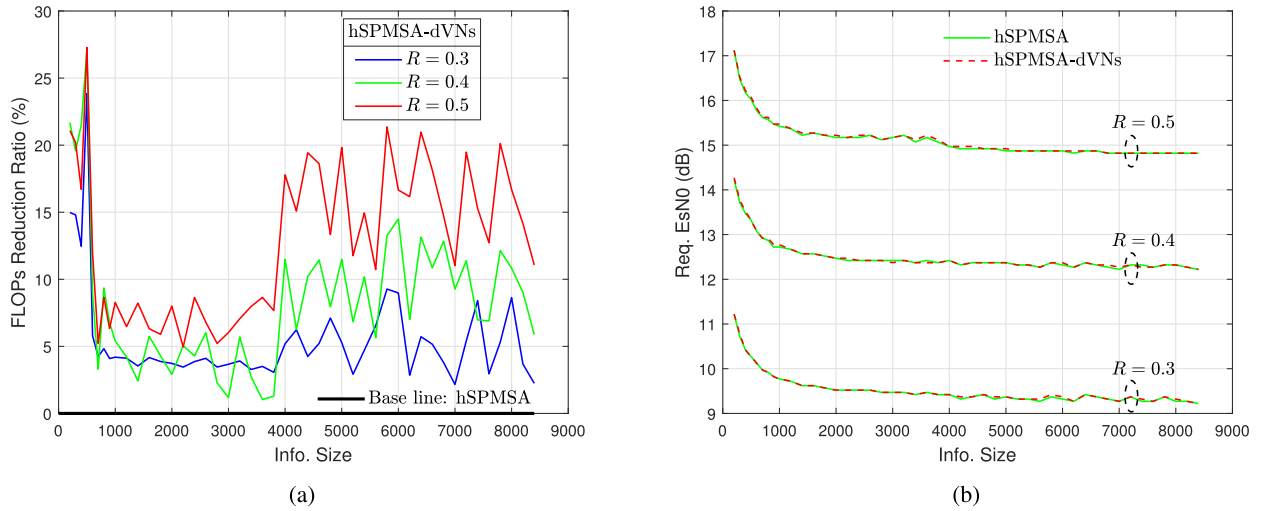


FIGURE 10. (a) Complexity and (b) performance comparison for QAM-256 between hSPMSA-dVNs with $P_d = 10^{-3}$ and hSPMSA. $n_{\text{SPA}} = 76$ for BG1 and 36 for BG2.

no performance loss. The design parameters P_d and n_{SPA} remain unchanged from the original specifications utilized during individual employment of dVNs and hSPMSA. The results illustrate that dVNs and hSPMSA can be combined straightforwardly without extra efforts like jointly designing P_d and n_{SPA} .

VI. CONCLUSION

We introduce two methods, namely dVNs and hSPMSA, to reduce the complexity of the layered decoding for QC-LDPC codes. dVNs can be enhanced by utilizing the unequal-error-protection provided by the bit-level channels in BICM systems. A variable node is deactivated when it is assigned to the most reliable bit-level channels and its initial LLR is larger than a threshold. The threshold is also carefully designed on the basis of the PDF of the initial LLRs.

The LLR updates regarding dVNs are omitted. In hSPMSA, we combine SPA and MSA by implementing them across different layers. A degree-based method is proposed to determine the layers employing SPA and MSA. Specifically, SPA tends to be employed by the low-degree layers, while MSA is applied to the remaining layers. In addition, we can incorporate dVNs and hSPMSA into the layered decoding together without any conflict to provide joint reduction in complexity. Through comprehensively evaluating 5G NR LDPC codes, the proposed methods demonstrate outstanding complexity-performance trade-off.

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HEPING WAN received the B.S. and M.S. degrees in telecommunication engineering from Xidian University, Shaanxi, China, in 2014 and 2017, respectively, and the Ph.D. degree in electrical engineering from The University of Texas at Dallas, Richardson, TX, USA, in 2022. Since 2022, he has been an Engineer with Samsung Research America, focusing on information theory, coding theory, and their applications.



JOONYOUNG CHO received the B.S., M.S., and Ph.D. degrees in electrical engineering from Pohang University of Science and Technology (POSTECH), South Korea, in 1993, 1995, and 2003, respectively. He engaged in CDMA baseband modem algorithm development and design, while he was with SK Telecom, South Korea, from 1995 to 1998, and a Research Staff with POSTECH, until 1999. He was also involved with research and standardization for 3G(HSPA)/4G(LTE)/5G(NR) cellular communications, while he was with Samsung Electronics, South Korea, from 2003 to 2015, and Intel Corporation, USA, from 2015 to 2018. He is currently the Senior Director of Samsung Research America (SRA), where he leads standardization and research for 5G/6G wireless communications.



MIN JANG (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in information and communication engineering from Sungkyunkwan University, Suwon, South Korea, in 2009, 2011, and 2015, respectively. Since 2015, he has been with Samsung Electronics, where he has been actively involved in the research, standardization, and development of wireless communication systems, including the fifth-generation (5G) mobile communications standards. He is currently a part of Samsung Research, Samsung Electronics, focusing on the research and development of the sixth-generation mobile communication standards. His research interests include coding theory, iterative information processing, communication systems, and AI-enabled communication technologies.



CHARLIE JIANZHONG ZHANG (Fellow, IEEE) received the Ph.D. degree from the University of Wisconsin, Madison. He is currently an EVP with Samsung Research America, where he leads research, prototyping, and standardization for 5G/6G and other wireless systems. He is also a Corporate VP and the Head of the Global 6G Team, Samsung Research. He is also the ATIS North America Next-G Alliance Full Member Group Vice Chair. He was the Board Chair of the FiRa Consortium from May 2019 to May 2023 and the Vice Chair of the 3GPP RAN1 Working Group from 2009 to 2013, where he led the development of LTE and LTE-advanced technologies. He received the 2024 IEEE ComSoc Industrial Innovation Award.

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